

PLANNING

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Identifying and Selecting System Development Projects:

The first phase of the SDLC is planning, consisting of project identification and selection, and project initiation and planning. During project identification and selection, a senior manager, a business group, an IS manager, or a steering committee identifies and assesses all possible systems development projects that an organization unit could undertake. Next, those projects deemed most likely to yield significant organizational benefits, given available resources are selected for subsequent development activities.

Information systems development requests come from a variety of sources. One source is requests from by managers and business units for replacing or extending an existing systems to gain needed information or to provide a new service to customers. Another source for requests is IS managers who want to make a system more efficient and less costly to operate, or want to move it to a new environment. A final source of projects is a formal planning group that identifies projects for improvements to help the organization meet its corporate objectives (e.g. a new system to provide better customer service).

PROCESS OF IDENTIFYING AND SELECTING IS DEVELOPMENT PROJECTS

Project identification and selection consists of three primary activities:

1. Identifying potential development projects
2. Classifying and ranking IS development projects
3. Selecting IS development projects.

1. Identifying potential development projects: Organizations vary as to how they identify projects. This process can be performed by

- a) A key member of top management, either the CEO of a small or medium sized organization or a senior executive in a larger organization.
- b) A steering committee, composed of a cross section of managers with an interest in systems.
- c) User departments, in which either the head of the requesting unit or a committee from the requesting department decides which projects to submit.
- d) The development group or a senior IS manager.

Characteristics of Alternative Methods for making IS identification and selection decisions:

* Top management:

- Greater strategic focus → Largest project size
- Longest project duration → Enterprise-wide considerations

* Steering Committee:

- Cross-functional focus → Greater Organizational change
- formal cost-benefit analysis → Larger and riskier projects.

* Functional Area:

- Narrow, non strategic focus → Faster development
- Fewer users, management layers, and business functions involved.

* Development Group

- Integration with existing systems flows
- fewer development delays
- Less concern with cost-benefit analysis.

Projects are identified by both top-down and bottom-up initiatives.

2. Classifying and ranking IS development projects:

Classifying and ranking projects can be performed by top managers, a steering committee, business units, or the IS development group. The criteria used to assign the merit of a given project can vary based on the size of the organization. The criteria are cost, duration, complexity, system size and focus.

Possible Evaluation Criteria when classifying and ranking projects:

- * Value Chain Analysis: Extent to which activities add value and costs when developing products and/or services.
- * Strategic Alignment: Extent to which the project is viewed as helping the organization achieve its strategic objectives and long-term goals.
- * Potential Benefits: Extent to which the project is viewed as improving profits, customer service, and so forth, and the duration of these benefits.

- * Resource Availability: Amount and type of resources the project requires and their availability.
- * Project size/duration: Number of individuals and the length of time needed to complete the project.
- * Technical Difficulty/Risks: Level of technical difficulty to successfully complete the project within given time and resource constraints.

Value chain analysis is an important project evaluation method that is widely used for assessing information systems development projects. It is the process of analyzing an organization's activities for making products and/or services to determine where value is added and costs are incurred. Once an organization gains a clear understanding of its value chain, improvements in the organization's operations and performance can be achieved.

3. Selecting IS development projects: The selection of projects is the final activity in the project identification and selection phase. Business objectives for both short- and long-term projects are considered. As business conditions change over time, the relative importance of any single project may substantially change. Thus the identification and selection of projects is an important and ongoing activity. Numerous factors must be considered when selecting a project. These factors include:
- Perceived needs of the organization
 - Existing systems and ongoing projects
 - Resource availability

- Evaluation criteria
- Current business conditions
- perspectives of the decision makers.

The decision-making process can lead to numerous outcomes.

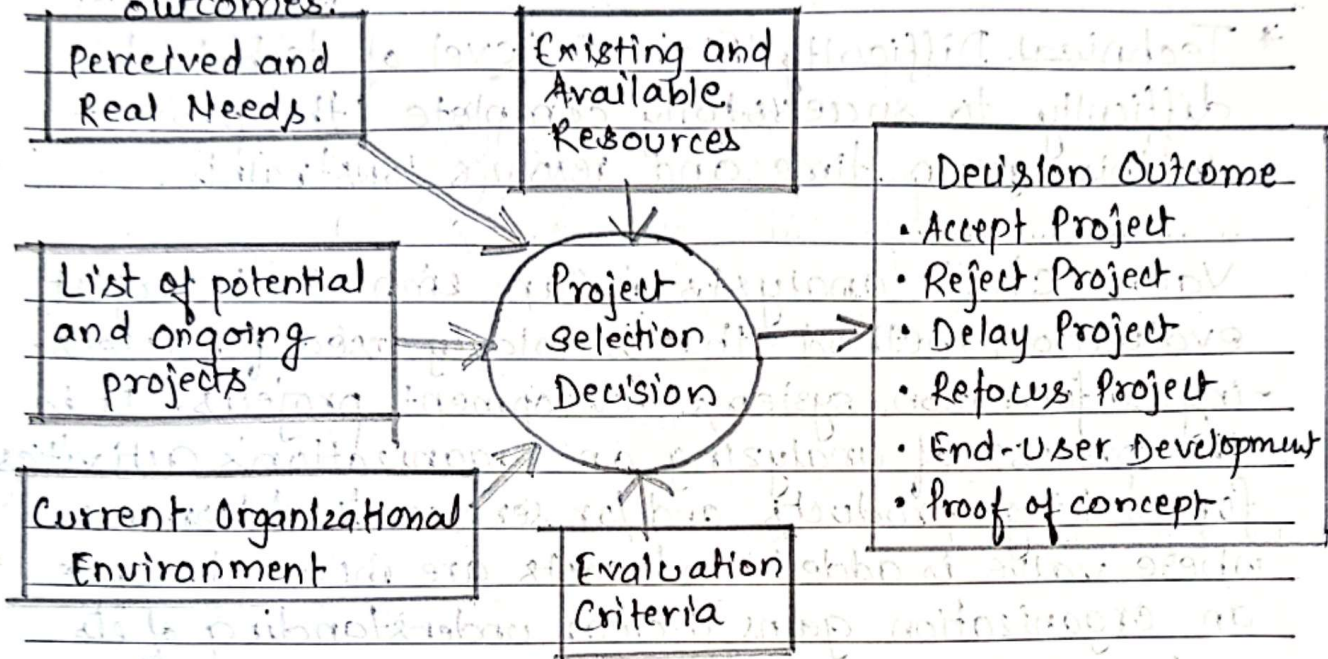


fig: Numerous factors and numerous outcomes for project selection decision.

Deliverables and Outcomes

The primary deliverable from the project identification and selection phase is a schedule of specific IS development projects. An outcome of this activity is the assurance that people in the organization gave careful consideration to project selection and clearly understood how each project could help the organization reach its objectives. Because of the principle of incremental commitment, a selected project does not necessarily result in a working system. Incremental commitment is a strategy in system analysis and design in which the project is reviewed after each

phase and continuation of the project is rejustified in each of these reviews.

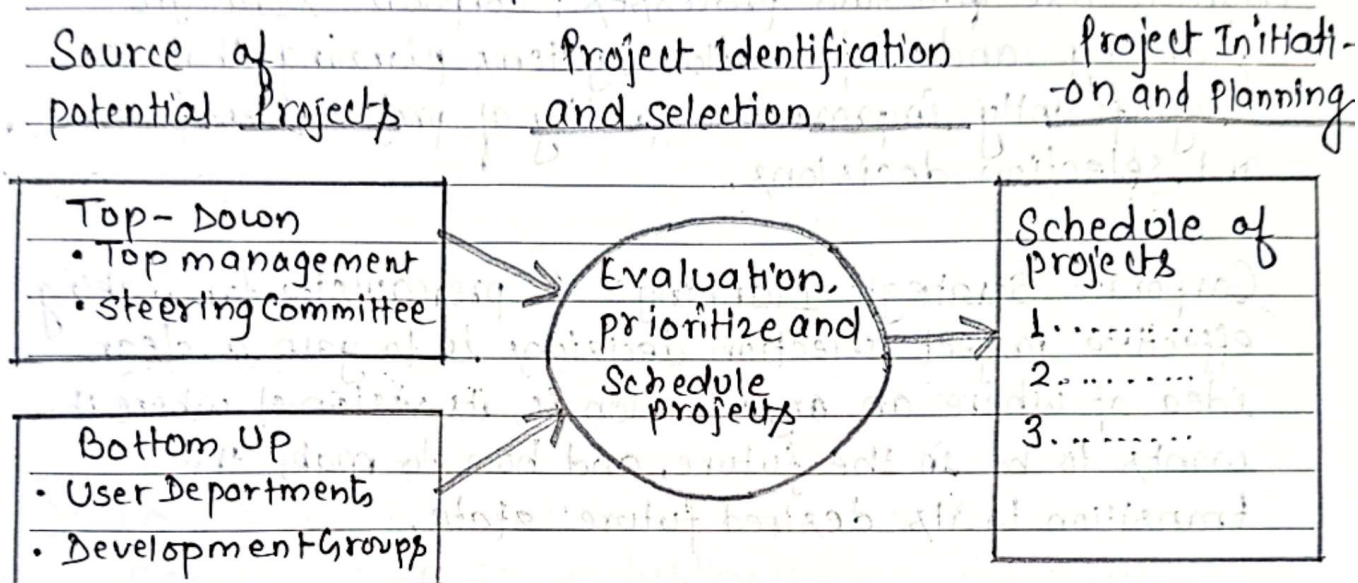


fig: IS development outcome from both top-down & Bottom up initiatives.

Corporate and Information Systems Planning

The need for improved information systems project identification and selection is readily apparent when we consider factors such as the following:

1. The cost of information systems has risen steadily and approaches 40 percent of total expenses in some organizations.
2. Many systems cannot handle applications that cross organizational boundaries.
3. Many systems often do not address the critical problems of the business as a whole or support strategic applications.
4. Data redundancy is often out of control, and users may have little confidence in the quality of data.
5. Systems maintenance costs are out of control as old, poorly planned systems must constantly be revised.

Careful planning and selection of projects alone will certainly not solve all of these problems. We can in addition use ~~for~~ two processes: corporate strategic planning and information systems planning that can significantly improve the quality of project identification and selection decisions.

Corporate Strategic Planning: A prerequisite for making effective project selection decisions is to gain a clear idea of where an organization is, its vision of where it wants to be in the future, and how to make the transition to its desired future state.

Step 1	Step 2	Step 3
Current Enterprise	Future Enterprise	Strategic Plan

fig.: Corporate strategic planning; a three-step process

The first step focuses on gaining an understanding of the current enterprise. Next, top management must determine where it wants the enterprise to be in the future. Finally, a strategic plan can be developed to guide this transition after understanding the current and future enterprise.

The process of developing and refining models of the current and future enterprise as well as a transition strategy is often referred to as corporate strategic planning. During corporate strategic planning, executives typically develop a mission statement, statements of future corporate objectives, and strategies designed to help the organization reach its objectives.

The mission statement of a company typically states in very simple terms what business the company is in.

Objective statements refers to "broad and timeless" goals for the organization. These goals can be expressed as a series of statements that are either qualitative or quantitative but that typically do not contain details likely to change substantially over time. A competitive strategy is the method by which an organization attempts to achieve its mission and objectives. In essence, the strategy is an organization's game plan for playing in the competitive business world.

Generic Competitive Strategies

- * Low cost Producer: This strategy reflects competing in an industry on the basis of product or service cost to the consumer.
- * Product Differentiation: This competitive strategy reflects capitalizing on a key product criterion requested by the market (for example, high quality, style, performance, roominess).
- * Product Focus or Niche: This strategy is similar to both the low-cost and differentiation strategies but with a much narrower market focus.

Summary: To build the most effective information systems, it is only through the clear understanding of organizational mission, objectives, and strategies that IS development projects should be identified and selected.

Information Systems Planning :

The second planning process that can play a significant role in the quality of project identification and selection decisions is called information systems planning (ISP). ISP is an orderly means of assessing the information needs of an organization and defining the information systems, databases and technologies that will best satisfy those needs.

The three key activities of this modeling process are :

1. Describing Current Situation
2. Describing the Target Situation
3. Developing a Transition Strategy and Plans.

1. Describing Current Situations:

- Listing of manual and automated processes
- Listing of manual and automated data
- Technology inventory
- Human resources inventory.

The most widely used approach for describing the current organizational situation is generically referred to as top-down planning.

Advantages of top-down planning over other approach

- Broader Perspective
- Improved Integration
- Improved Management Support
- Better Understanding

2. Describing the Target Situation, trends, and Constraints :

- Blueprints of manual and automated processes
- Blueprints of manual and automated data.
- Technology blueprints
- Human resources blueprints.

Next step in ISP is to define the target situation that reflects the desired future state of organization. Target situation consists of desired state of locations, units, functions, processes, data, IS, trends of the organization environment (e.g, time, resources, technological evolution, competition and so, on). Planners then focus on the differences between the current and future lists and matrices to identify projects and transition strategies.

3. Developing a Transition Strategy and Plans.

→ Schedule of Projects.

Once the creation of the current and target situations is complete, a detailed transition strategy and plan are developed by the IS planning team. This plan should be very comprehensive, reflecting broad, long-range issues in addition to providing sufficient detail to guide all levels of management concerning what needs to be done, how, when and by whom in organization.

The components of a typical ISP :-

- * Organizational Mission, Objectives, and Strategy:
Briefly describes the mission, objectives and strategy of the organization. The current and future views of the company are also briefly presented.
- * Informational Inventory:
This section provides a summary of the various business processes, functions, data entities, and information needs of the enterprise. This inventory will view both current and future needs.
- * Mission and Objectives of IS:
Description of the primary role IS will play in the organization to transform the enterprise from its current to future state.
- * Constraints on IS Development:
Briefly describes limitations imposed by technology and current level of resources within the company - financial, technological and personnel.
- * Long-Range IS Strategies:
Presents a summary of the overall systems needed within the company and the set of long-range (2-5 years) strategies chosen by the IS department to fill the needs.
- * Short-Term Plan:
Shows a detailed inventory of present projects and systems and a detailed plan of projects to be developed or advanced during the current year.

* Conclusions: Contains likely but not-yet certain events that may affect the plan, an inventory of business change elements as presently known and a description of their estimated impact on the plan.

INITIATING AND PLANNING SYSTEMS DEVELOPMENT PROJECTS

Project identification and selection is often thought of as a "preproject" step in the life cycle. This recognition of potential projects may come as part of a larger planning process, information systems planning, or from requests from managers and business units.

Regardless of how a project is identified and selected, the next step is to conduct a more detailed assessment during project initiating and planning. This assessment does not focus on how the proposed system will operate but rather on understanding the scope of a proposed project and its feasibility of completion given the available resources. It is crucial that organizations understand whether resources should be devoted to a project; otherwise very expensive mistakes can be made.

Many activities performed during initiation and planning could also be completed during the next-phase of the SDLC - systems analysis. Proper and insightful project initiation and planning, including determining project scope and identifying project activities, can reduce the time needed to complete later project phases, including systems analysis.

Most organizations assign an experienced system analyst, or a team of analysts, for large projects, to perform project initiation and planning. The objective

of project initiating and planning is to transform a unclear system request document into a tangible project. Effective communication among the systems: Analysts, Users, and management is essential for the creation of meaningful project plan.

Process of Initiating and Planning IS Development PROJECTS

Two major activities occur during project initiation and project planning. Project initiation focuses on activities that will help organize a team to conduct project planning. During initiation, one or more analysts are assigned to work with a customer to establish work standards and communication procedures.

Elements of project Initiation:

- Establishing the project initiation team.
- Establishing a Relationship with the customer
- Establishing the Project Initiation Plan.
- Establishing Management procedures.
- Establishing the project management environment and project Workbook.
- Developing the project charter.

The second activity, project planning, focuses on defining clear, discrete tasks and the work needed to complete each task. The objective of the project planning process is to produce two documents: a baseline project plan (BPP) and the project scope statement (PSS). The BPP becomes the foundation for the remainder of the development project. The PSS, produced by the project team, clearly outlines the objectives of the

project for the customer.

Elements of Project Planning:

- Describing the project scope, Alternatives, and Feasibility.
- Dividing the Project into Manageable Tasks
- Estimating Resources and Creating a Resource Plan.
- Developing a Preliminary Schedule.
- Developing a Communication Plan.
- Determining Project Standards and Procedures
- Identifying and Assessing Risk.
- Creating a Preliminary Budget.
- Developing the Project Scope Statement
- Setting a Baseline Project Plan.

Deliverables and Outcomes

The major outcomes and deliverables from project initiation and planning are the baseline project plan and the project scope statement.

The baseline project plan (BPP) contains all information collected and analyzed during the project initiation and planning activity. The plan contains the best estimate of the project's scope, benefits, costs, risks and resources requirements given the current understanding of the project. BPP is used to help decide whether to continue, redirect, or cancel a project. If selected, the BPP becomes the foundation document for all subsequent SDLC activities.

The Project Scope Statement (PSS) is a short document prepared for the customer that describes what the project will deliver and outlines all work required to complete the project. PSS typically consists of a high level summary of the BPP information.

ASSESSING PROJECT FEASIBILITY

A feasibility study is an analysis that considers all of a project's relevant factors including economic, technical, legal and scheduling considerations to ascertain the likelihood of completing the project successfully. Whether a project is feasible or not can depend on several factors, including the project's cost and return on investment.

Importance

- It helps to define our goals and objectives.
- It helps us to develop a plan.
- It helps to execute the developed plan.
- It gives us an identity.
- It improves the attention of project teams.
- Reduces the number of business options.

Types of Project Feasibility

1) Economic Feasibility / Cost-benefit analysis:

Economic feasibility or cost-benefit analysis is the analysis of system to identify the financial benefits and costs associated with the development project. If the expected benefit equal or exceed the expected costs.

During project initiation and planning, it will be impossible for us to define precisely all benefits and costs related to a particular project. We review worksheets to record costs and benefits. These worksheets are used after each SDLC phase to decide whether to continue, redirect, or kill the project.

2) Technical Feasibility:

Assessing technical feasibility is to evaluate whether the new system will perform adequately and whether an organization has ability to construct a proposed system or not. All projects have risk and that risk is not necessarily something to avoid. Organizations typically expect a greater return on their investment for riskier projects. Potential risks need to be identified as early as possible in their project.

The amount of technical risk associated with a given project depends on four primary factors: project size, project structure, the development group's experience with the application and technology area, and the user group's experience with systems development projects and the application area.

3) Operational Feasibility:

Assessing operational feasibility is to gain an understanding of whether the proposed system will likely to solve the business problems, or take advantage of opportunities or not. It is important to understand how the new systems will fit into the current day-to-day operations of the organization.

It is not only important to evaluate whether a system can work, but also evaluate whether a system will work. A workable solution might fail because of the end-user. Such as how long will the working environment of the end-users change, or whether end-users will adopt to that change.

4) Schedule Feasibility :

Assessing schedule feasibility is to access the duration of the project whether it is too long to be complete before it is useful. System analysts have to estimate how long the system will take to develop.

Furthermore the learning curve of the new technology and new system should be considered. We may have the technology, but that doesn't mean we have the skills required to properly apply that technology. Even though all information system professionals can learn new technologies, the learning curve will specifically impact the schedule feasibility.

5) Legal and Contractual Feasibility :

Legal feasibility determines whether the proposed system conflicts with the legal requirement or not. A project may face legal issues after the completion if this factor is not considered at the first stage. The possible considerations might include copyright, labour laws, creation of data to share data with other organizations etc.

6) Political Feasibility :

Assessing political feasibility is to gain an understanding of how key stakeholders within the organization view the proposed system. Information system may affect the distribution of information within the organization and thus the distribution of power. Those stakeholders not supporting the project may take steps to block, disrupt or change the intended focus of the project.

Steps of Cost-benefit analysis:

1) Determining Project Benefits

Project Benefits are both tangible and intangible. Tangible benefits refer to items that can be measured in dollars and with certainty. All tangible benefits can be easily quantified. Most tangible benefits will fit within the following categories:

- cost reduction and avoidance
- Error reduction
- Increased flexibility
- Increased speed of activity
- Improvement of management planning and control.
- Opening new markets and increasing sales opportunities.

Intangible benefits refer to item that cannot be easily measured in dollars and with certainty. These benefits may have direct organizational benefits, such as the improvement of employee morale, or they may have broader societal implications such as reduction of waste creation or resource consumption. Potential tangible benefits may have to be considered intangible during PIP because we may not be able to quantify them in dollars or with certainty at this stage. During later stages such intangibles can become tangible benefits as we better understand the ramifications of the system being designed.

Some Intangible Benefits :

- Competitive necessity
- More timely information
- Improved organizational planning
- Increased organizational flexibility
- Foster decision making
- Improved processing efficiency
- Improved asset utilization
- Improved resource control
- Positive impacts on society
- Improved social responsibility
- Better usage of resources.

27 Determining Project Costs:

An information system can have both tangible and intangible costs. Tangible costs include items such as hardware costs, labor costs and operational costs including employee training and renovations. Intangible costs can include loss of customer goodwill, employee morale or operational inefficiency. The goal of a cost-benefit analysis is to accurately determine the total cost of ownership (TCO) for an investment.

Besides tangible and intangible costs, IS related development costs can be distinguished as either one-time cost or recurring cost.

A one-time cost refers to a cost associated with project initiation and development and the start-up of the system. ~~Recurring~~ examples: cost associated to system development, new hardware and software purchase, user training, site preparation and data or system conversion.

Recurring costs refers to those costs resulting from the ongoing evolution and use of the system. Examples of these costs typically include:

- Application software maintenance
- Incremental data storage expenses
- Incremental communications
- New software and hardware leases.
- Supplies and other expenses (e.g. paper, forms etc)

Commonly used Cost-Benefit Analysis techniques:

1) Net Present Value (NPV): Present value is the current value of a dollar at any time in the future. NPV uses a discount rate determined from the company's cost of capital to establish the present value of a project.

Present value is calculated using formula:

$$PV_n = Y \times \frac{1}{(1+i)^n}$$

where, PV_n is the present value of Y dollars in n years from now, i is the discount rate, The rate at which money can be borrowed or invested.

To calculate NPV, we simply add the present values calculated previously.

2) Return on Investment (ROI): ROI is the ratio of the net cash receipts of the project divided by the cash outlays of the project. Trade-off analysis can be made among projects competing for investment by comparing their representative ROI ratios. Firms choose the project that provides the highest ROI.

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Overall return on investment (ROI) = $\frac{\text{Overall NPV}}{\text{NPV of all costs.}}$

3) Break-Even Analysis (BEA): BEA finds the amount of time required for the cumulative cash flow from a project to equal its initial and ongoing investment.

Break-even ratio = $\frac{\text{Yearly NPV cash flow} - \text{Overall cash flow}}{\text{Yearly NPV cash flow}}$

The break-even point is usually expressed as the amount of revenue that must be realized for the firm to have neither profit or loss. It uses a minimum revenue target. It can be expressed in numbers or by use of graphs.

Numerical Example

Assuming monetary benefits of an information system at \$85,000 per year, one-time cost of \$75,000, recurring costs of \$35,000 per year, a discount rate of 12 percent, and a five-year time horizon, calculate the net present value of these costs and benefits of an information system. Also calculate the overall return on investment of the project and then present a break analysis. At what point does break even occur?

Solution

Given, Monetary benefits of an IS = \$85,000 per year

One-time costs = \$75,000

Recurring costs = \$35,000 per year

Discount rate = 12%

Time period = 5 years.

Present Value of Benefits

Since Benefits start from year 1, so.

$$PV_0 = 0$$

$$PV_1 = 85000 \times \frac{1}{(1+0.12)^1} = 75,893$$

$$PV_2 = 85000 \times \frac{1}{(1+0.12)^2} = 67,761$$

$$PV_3 = 85000 \times \frac{1}{(1+0.12)^3} = 60,501$$

$$PV_4 = 85000 \times \frac{1}{(1+0.12)^4} = 54,019$$

$$PV_5 = 85000 \times \frac{1}{(1+0.12)^5} = 48,231$$

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Now, Net Present Value of Benefits (NPV) = $PV_0 + PV_1 + PV_2 + PV_3 + PV_4 + PV_5$

$$= 75,893 + 67761 + 60501 + 54019 + 48231$$

$$= \$306405$$

Present value of costs

Here, one-time cost \$75000 is treated as cost occurring in year 0.

$$PV_0 = 75000 \times \frac{1}{(1+0.12)^0} = \$75000$$

Recurring cost \$35000 happens every year, starting at year 1.

$$PV_1 = 35000 \times \frac{1}{(1+0.12)^1} = 31,250$$

$$PV_2 = 35000 \times \frac{1}{(1+0.12)^2} = 27901$$

$$PV_3 = 35000 \times \frac{1}{(1+0.12)^3} = 24912$$

$$PV_4 = 35000 \times \frac{1}{(1+0.12)^4} = 22243$$

$$PV_5 = 35000 \times \frac{1}{(1+0.12)^5} = 19860$$

$$\begin{aligned} \text{NPV of costs} &= PV_0 + PV_1 + PV_2 + PV_3 + PV_4 + PV_5 \\ &= 75000 + 31250 + 27901 + 24912 + 22243 + 19860 \\ &= \$201168 \end{aligned}$$

for Overall Return on Investment

$$\begin{aligned} \text{Overall NPV} &= \text{NPV of all Benefits} - \text{NPV of all costs} \\ &= \$306405 - \$201168 \\ &= \$105239 \end{aligned}$$

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$$\therefore \text{Overall ROI} = \frac{\text{Overall NPV}}{\text{NPV of all costs}} = \frac{105239}{201168} = 0.523$$

For Break-even ratio:

$$\text{Break-even ratio} = \frac{\text{Yearly NPV cash flow} - \text{Overall NPV cash flow}}{\text{Yearly NPV cash flow}}$$

So, before finding BER, we need to calculate year in which break even occurs and yearly NPV cash flow and overall NPV cash flow.

$$\text{Yearly NPV cash flow} = \text{PV of Benefits} - \text{P.V. of Recurring costs.}$$

$$\text{PV}_0 \text{ of benefit} = 0.$$

$$\text{yearly NPV cash flow at year 0} = -75000$$

$$\text{yearly NPV cash flow at year 1} = \text{PV}_1 \text{ of benefit} - \text{PV}_1 \text{ of cost.}$$

$$= 75893 - 31250$$

$$= 44643$$

Similarly,

$$\text{yearly NPV cash flow at year 2} = 67761 - 27901 = 39860$$

$$\text{yearly NPV cash flow at year 3} = 60501 - 24912 = 35589$$

$$\text{yearly NPV cash flow at year 4} = 54019 - 22243 = 31776$$

$$\text{yearly NPV cash flow at year 5} = 48231 - 19860 = 28371$$

Since overall NPV cash flow for i th year

$$= \text{yearly NPV cash flow for } i\text{th year} + \text{yearly overall NPV cash flow for } (i-1)\text{th year.}$$

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Overall NPV cash flow for 0th year = -75000Overall NPV cash flow for 1st year = ~~44643~~

$$44643 + (-75000)$$

$$= -30357$$

Overall NPV cash flow for 2nd year = $39860 + (-30357)$

$$= 9503$$

Overall NPV cash flow for 3rd year = $35589 + 9503$

$$= 45092$$

Overall NPV cash flow for 4th year = $31776 + 45092$

$$= 76868$$

Overall NPV cash flow for 5th year = $28371 + 76868$

$$= 105239$$

Here break-even occurs between year 1 and 2

because year 2 is first in which overall NPV cash flow is non-negative.

So, break-even ratio = $\frac{\text{yearly cash flow at year 2} - \text{Overall NPV cash flow for 1st year}}{\text{yearly cash flow at year 2}}$

$$= \frac{39860 - (-30357)}{39860} = 0.76$$

∴ Project ~~even~~ break-even occurs at approximately in 1.76 year

Note: project even is written as lower bound year when break-even occurs + breakeven ratio.

** We can solve this making table

Cost Benefit Analysis							
	year of project						
	year 0	year 1	year 2	year 3	year 4	year 5	TOTALS
Net Benefit	\$ -	\$ 85,000	\$ 85,000	\$ 85,000	\$ 85,000	\$ 85,000	
Discount rate (12%)	1.0000	0.8929	0.7972	0.7118	0.6355	0.5674	
PV of Benefits	\$ -	\$ 75,892.86	\$ 67,761.48	\$ 60,501.32	\$ 54,019.04	\$ 48,231.28	
NPV of All benefits	\$ -	\$ 75,892.86	\$ 143,654.34	\$ 204,155.66	\$ 258,174.69	\$ 306,405.98	\$ 306,405.98
one time cost	\$ (75,000.0)						
recurring cost	\$ -	\$ (35,000.00)	\$ (35,000.00)	\$ (35,000.00)	\$ (35,000.00)	\$ (35,000.00)	
discount rate(12%)	1.0000	0.8929	0.7972	0.7118	0.6355	0.5674	
PV of Recurring cost	\$ -	\$ (31,250.00)	\$ (27,901.79)	\$ (24,912.31)	\$ (22,243.13)	\$ (19,859.94)	
NPV of all costs	\$ (75,000.00)	\$ (106,250.00)	\$ (134,151.79)	\$ (159,064.09)	\$ (181,307.23)	\$ (201,167.17)	\$ (201,167.17)
overall NPV							\$ 105,238.81
overall ROI-(overall NPV-NPV of ALL COST)							0.523141085
BREAK EVEN ANALYSIS							
YEARLY NPV CASH FLOW	\$ (75,000.00)	\$ 44,642.86	\$ 39,859.69	\$ 35,589.01	\$ 31,775.90	\$ 28,371.34	
OVERALL NPV CASH FLOW	\$ (75,000.00)	\$ (30,357.14)	\$ 9,502.55	\$ 45,091.56	\$ 76,867.47	\$ 105,238.81	
project break even occurs between years 1 and 2							
break even ratio=							0.7616
Actual break even occurred at 1.7 years							

Building and Reviewing the Baseline Project Plan

A baseline in project management is clearly defined starting point for our project plan. It is a fixed reference point to measure and compare our project's progress. This allows us to assess the performance of our project over time.

Building Project Baseline:

To set a project baseline, first we must determine the scope of our project. This can be done with a scope statement which is a list of project objectives and deliverables. Now, we break down all the works required to achieve these deliverables into individual tasks and subtasks with detailed description.

Next we must map out our project schedule, clearly defined due dates and a final deadline. A good tool for this is Gantt chart. After that, we should plan the total costs of project. It includes relevant factors like hourly rates, available resources etc. Finally project baseline must be clearly outlined before we can move to next step or phase.

Outline of a baseline project plan

BASELINE PROJECT PLAN REPORT

I. INTRODUCTION

A. Project Overview - Provides an executive summary that specifies the project's scope, feasibility, justification, resource requirements, and schedules. Additionally, a brief statement of the problem, the environment in which the system is to be implemented, and constraints that affect the project.

are provided.

- B. Recommendation - Provides a summary of important findings from the planning process and recommendations for subsequent activities.

2. SYSTEM DESCRIPTION

- A. Alternatives - Provides a brief presentation of alternative system configurations.
- B. System Description - Provides a description of the selected configuration and a narrative of input information, tasks performed, and resultant information.

3. FEASIBILITY ASSESSMENT

- A. Economic Analysis - Provides an economic justification for the system using cost-benefit analysis.
- B. Technical Analysis - Provides a discussion of relevant technical risk factors and an overall risk rating of the project.
- C. Operational Analysis - Provides an analysis of how the proposed system solves business problems or takes advantage of business opportunities in addition to an assessment of how current day-to-day activities will be changed by the system.
- D. Legal and Contractual Analysis - Provides a description of any legal or contractual risks related to the project.
- E. Political Analysis - Provides a description of how key stakeholders within the organization view the proposed system.
- F. Schedules, Time Line, and Resource Analysis -

provides a description of potential time frame and completion date scenarios using various resource allocation schemes.

4. MANAGEMENT ISSUES

- A. Team Configuration and Management - Provides a description of the team member roles and reporting relationships.
- B. Communication Plan - Provides a description of the communication procedures to be followed by management, team members, and the customer.
- C. Project Standards and Procedures - Provides a description of how deliverables will be evaluated and accepted by the customer.
- D. Other Project-specific Topics - Provides a description of any other relevant issues related to the project uncovered during planning.

Reviewing Projects Baseline

After building project baseline, management, users and development team will review the document. A project baseline plan should not ~~be~~ change without following formal change control procedures.

A common method for performing review is called a structured walk-through. Walk-throughs are peer group reviews of any product created during the system's development process and are widely used by professional development organizations. At walkthrough meetings, there is need to have individuals play specific roles. These roles are as follows:

- * Coordinator: This person plans the meeting and facilitates a smooth meeting process. This person may be the project leader or a lead analyst responsible for the current life cycle step.
- * Presenter: This person describes the work product to the group. The presenter is usually an analyst who has done all or some of the work being presented.
- * User: This person (or group) makes sure that the work product meets the needs of the project's customers. This user would usually be someone not on the project team.
- * Secretary: This person takes notes and records decisions or recommendations made by the group. This person may be one of the analyst or clerk assigned to the project.
- * Standards bearer: The role of this person is to ensure that the work product adheres to organizational technical standards. Many larger organizations have staff groups within the unit responsible for establishing standard procedures, methods and documentation formats. These standards bearer validate the work so that it can be used by others in the development organization.
- * Maintenance Oracle: This person reviews the work product in terms of future maintenance activities. The goal is to make the system and its documentation easy to maintain.